Ninth assignment

Remote Group

For November 15th

Most of the following meerely qualify as proof sketches, and (e) is missing. As I will be away from now until the deadline on sunday I leave it as is.

Problem (Munkres 55.4). Suppose you are given the fact that for each n, there is no retraction $r : B^{n+1} \to S^n$. Prove the following:

(a) The identity map $i : S^n \to S^n$ is not nullhomotopic.
(b) The inclusion map $j : S^n \to \mathbb{R}^{n+1} - 0$ is not nullhomotopic.
(c) Every nonvanishing vector field on B^{n+1} points directly outward at some point of S^n, and directly inward at some point of S^n.
(d) Every continuous map $f : B^{n+1} \to B^{n+1}$ has a fixed point.
(e) Every $n+1$ by $n+1$ matrix with positive real entries has a positive eigenvalue.
(f) If $h : S^n \to S^n$ is nullhomotopic, then h has a fixed point and h maps some point x to its antipode $-x$.

Proof of (a). We will prove by contrapositive.

Apologies if there are some slips in the rigor here, this was hard.

Suppose the identity $i : S^n \to S^n$ is nullhomotopic.

Then there is a continuous map

$$H : S^n \times I \to S^n$$

such that

$$H(x, 0) = x, \; H(x, 1) = p$$

for some fixed $p \in S^n$.

Starting with $S^n \times I$ we can impose $(x, 1) \sim (y, 1)$ for all $x, y \in S^n$. H is then constant on all equivalence classes, letting us factor

$$\begin{array}{ccc} S^n \times I & \xrightarrow{H} & S^n \\ & \searrow^{q} & \uparrow{\scriptstyle \varphi} \\ & & (S^n \times I)/(\sim) \end{array}$$

giving a continuous map φ for which the diagram commutes (q is $q(x, t) = [(x, t)]$).

For this map we have that

$$\varphi\big|_{S^n\times 0} = H\big|_{S^n} = \mathrm{id}_S^n.$$

Now it remains to show that

$$(S^n \times I)/\sim \cong B^{n+1}.$$

Take $f : (S^n \times I)/\sim \longrightarrow B^{n+1}$ by

$$f([(x_1, x_2, \dots, x_{n+1}), t]) = (1-t)(x_1, x_2, \dots, x_{n+1}).$$

Then the point p corresponding to the nontrivial equivalence class is the origin.

The map is a well-defined homeomorphism (I swear!).

This gives a retraction map from $B^{n+1} \to S^n$ which we were told wasn't possible! ■

Proof of (b). I'm sure one could argue this from the fact that $S^n \simeq \mathbb{R}^{n+1} \setminus 0$, but instead, take the retraction

$$r : \mathbb{R}^{n+1} \setminus 0 \to S^n,\ r(x) = \frac{x}{\|x\|}.$$

Observe that

$$r \circ j = \mathrm{id}_{S^n}.$$

Suppose, for contradiction, that j is nullhomotopic. Then

$$j \simeq c_p$$

for some constant map $c_p : S^n \to \mathbb{R}^{n+1} \setminus 0$.

Composing with r, we get

$$\mathrm{id}_{S^n} = r \circ j \simeq r \circ c_p,$$

where $r \circ c_p$ is constant, contradicting (a). ■

Proof of (c). A nonvanishing vector field on B^{n+1} is a continuous map

$$v : B^{n+1} \to \mathbb{R}^{n+1}$$

such that

$$v(x) \neq 0\ \forall x \in B^{n+1}.$$

Since v is nonvanishing,

$$u(x) = \frac{v(x)}{\|v(x)\|} : B^{n+1} \to S^n$$

is well-defined and continuous.

Suppose v never points directly outward. Then

$$u(x) \neq x \text{ for all } x \in S^n.$$

Something something... ■

Proof of (d). Suppose $f : B^{n+1} \to B^{n+1}$ has no fixed point, meaning

$$f(x) \neq x, \ \forall x \in B^{n+1}.$$

Intuitively, if we draw a ray from $f(x)$ to x and extend in far enough we reach the boundary S^n. Such a map is well-defined since $f(x) \neq x$.

Specifically we can formalize a ray

$$\gamma_x(t)x + t(x - f(x)), \ t \geq 0.$$

Since $x - f(x)$ is nonzero, there is some t such that

$$\|\gamma(t)\| = 1.$$

We find t by

$$\begin{aligned} 1^2 &= \|x + t_x(x - f(x))\|^2 \\ &= \|x\|^2 + 2t_x\langle x, x - f(x)\rangle + t_x^2\|x - f(x)\|^2. \end{aligned}$$

Set $y = t_x$ to get

$$\|x - f(x)\|^2 \cdot y^2 + 2\langle x, x - f(x)\rangle \cdot y + \|x\|^2$$

which we can solve by the quadratic formula (and get some nonnegative answer since the root will be nonnegative, I think). Call the value we get τ_x.

The point being that we get a continuous map

$$r : B^{n+1} \to S^n, \ r(x) = x + \tau_x(x - f(x))$$

for which $r\,\big|_{S^n} = \mathrm{id}_{S^n}$ which we have been told is impossible!

■

Proof of (f). Suppose h is nullhomotopic.

We have that there is a map

$$H : S^n \times I \to S^n$$

for which

$$H(x, 0) = h(x), \ H(x, 1) = p$$

for some $p \in S^n$.

We obtain, as in (a), an extension $\tilde{h} : B^{n+1} \to S^n$ such that

$$\tilde{h}\,\big|_{S^n} = h.$$

Regard $\tilde{h}$ as a self-map of B^{n+1} with $\operatorname{im}\tilde{h} = S^n$, whcih is perfectly ok since $B^{n+1} \supseteq S^n$.

By (d) there is $x \in B^{n+1}$ such that $\tilde{h}(x) = x$, but $\tilde{h}(x) \in S^n$ so necessarily $x \in S^n$,

meaninig

$$h(x) = \tilde{h}(x) = x.$$

Now $-\tilde{h}$ is a continuous self map and again there is $y \in B^{n+1}$ such that

$$-\tilde{h}(y) = y$$

and since $-\tilde{h}(y) \in S^n$ we have $y \in S^n$ so

$$h(y) = \tilde{(y)} = -y.$$